Linux Networking Lab

Explanation Guide: what the commands in the README do

The README file lists every command. This guide explains what each group of commands is for, what you should see when you run it, and the idea it teaches. Read a section here, then run the matching commands from the README in Kali Linux.

1. Starting your lab

The lab runs inside Docker containers, so you can practise real protocols safely on your own machine. `docker compose up -d` builds and starts the services in the background. `docker compose ps` shows which containers are running and which ports they expose. `docker compose logs` prints what the services have been doing, which is useful when something does not work.

`docker compose down` stops and removes the containers, and `docker compose restart` restarts them without rebuilding.

- The lab services listen on high ports (8080, 2121, 2222) instead of the standard ones (80, 21, 22) so they do not clash with services already on your machine.

2. Basic Linux networking commands

These commands tell you about your own machine's network setup. `ip addr` and `ifconfig` show your network interfaces and IP addresses. `ip route` shows where traffic is sent, including your default gateway. `hostname` and `hostname -I` show your machine's name and IP addresses.

`ss -tulnp`, `ss -lnt` and `netstat -tulnp` list ports that are listening for connections. In `-tulnp`: t = TCP, u = UDP, l = listening, n = numeric ports, p = the process using the port. `ping 127.0.0.1` tests the loopback address; a reply means your network stack works.

The install line adds the tools used later: curl and wget (web), netcat (raw connections), telnet, ftp, nmap (scanning) and the SSH client.

- 127.0.0.1 (localhost) always points to your own machine. The lab runs there, so no traffic leaves your computer.

3. Nmap: service discovery

Nmap scans a target to find open ports. `nmap 127.0.0.1` checks the most common ports. `-sT` uses a full TCP connect scan, which completes the handshake with each port. `-sV` also probes each open port to identify the service and its version. `-p 8080` limits the scan to one port, which is faster and cleaner.

Expect to see ports 8080, 2121 and 2222 reported as open, with names such as http, ftp and ssh.

- Scanning only systems you own or have permission to test is essential. Here the target is your own lab.

4. HTTP: port 80 / 8080

HTTP is how browsers and web servers talk. `curl http://127.0.0.1:8080` fetches the web page and prints its HTML. `curl -I` sends a HEAD request and shows only the response headers (status code, server type, content type). `curl -v` shows the full conversation, including the request your machine sent and the response it got. `wget` downloads a page and saves it as a file.

The Netcat examples send HTTP by hand. The text `GET / HTTP/1.1` followed by a Host header and a blank line is exactly what a browser sends. Typing it yourself shows that HTTP is just readable text over a TCP connection. `HEAD` works the same way but the server returns headers only.

`curl http://127.0.0.1:8080/flag.html` requests a specific page. Look for the status code: 200 means found, 404 means not found.

- HTTP is plain text, so anyone on the network path could read it. That is why HTTPS exists.

5. FTP: port 21 / 2121

FTP transfers files between a client and a server. You log in with the lab credentials, then use commands inside the FTP prompt.

`pwd` shows your current remote folder, `ls` and `dir` list files, `status` shows the session settings, and `help` lists available commands. `get README.txt` downloads a file to your machine. `bye`, `quit` and `close` end the session or connection. `cd`, `mkdir` and `delete` change folders and files on the server and may be refused if the lab account has limited permission.

`ascii` and `binary` choose the transfer mode. ASCII is for text files and adjusts line endings. Binary copies bytes exactly and should be used for images, archives and programs.

- FTP sends the username and password in plain text. It is useful for learning, but SFTP or SSH should replace it in real use.

6. SSH: port 22 / 2222

SSH gives you an encrypted remote terminal. `ssh -p 2222 student@127.0.0.1` connects as user student on port 2222. The `-p` option is needed because the lab does not use the default port 22. The first time, SSH asks you to trust the server's fingerprint; this protects against connecting to an impostor.

Inside the session, `whoami` shows your user, `hostname` shows the server's name (different from your Kali machine, proving you are inside the container), `pwd` and `ls -la` show where you are and what is there, and `exit` closes the session.

- Unlike Telnet and FTP, everything in SSH, including the password, is encrypted.

7. Telnet: port 23

This section uses two terminals. In Terminal 1, `nc -lv 127.0.0.1 2323` makes Netcat listen on port 2323 (l = listen, v = verbose). In Terminal 2, `telnet 127.0.0.1 2323` connects to it. Anything typed in one terminal appears in the other.

To leave Telnet, press Ctrl and] together, then type `quit`.

- This is a plain TCP connection, not a real Telnet server. It shows the basic idea: two programs exchanging raw text over a socket. Real Telnet is unencrypted and has been replaced by SSH.

8. SMTP: port 25

SMTP is the protocol that sends email between servers. `ss -tulnp | grep :25` checks whether anything on your machine is listening on port 25; normally nothing is.

The commands listed are what a mail client says to a server. `EHLO` and `HELO` introduce the client. `MAIL FROM` names the sender. `RCPT TO` names the recipient. `DATA` starts the message body (ended by a line with a single dot). `QUIT` closes the session.

- These are SMTP protocol commands, not shell commands. You cannot run them in the terminal; they need an SMTP server on the other end, and the lab does not include one.

9. POP3: port 110

POP3 lets a mail client download messages from a mailbox. `USER` and `PASS` log in. `STAT` gives the number of messages and total size. `LIST` lists messages. `RETR 1` retrieves message number 1. `QUIT` ends the session.

- POP3 typically downloads mail to one device and may remove it from the server.

10. IMAP: port 143

IMAP also reads email, but keeps the mail on the server so many devices stay in sync. Each command starts with a tag (a1, a2, ...) so the server's reply can be matched to the request. `LOGIN` signs in, `LIST` shows mailboxes, `SELECT INBOX` opens the inbox, `STATUS` reports counts such as the number of messages, and `LOGOUT` ends the session.

- POP3 and IMAP commands are for understanding. The lab has no mail server, so you study them rather than run them.

11. HTTPS: port 443

HTTPS is HTTP wrapped in TLS encryption. `openssl s_client -connect example.com:443 -servername example.com` performs the TLS handshake and prints the certificate chain, the protocol version and the cipher used. `curl -Iv https://example.com` shows the same handshake in summary form, followed by the response headers.

Remember the standard ports: HTTP 80, HTTPS 443, SSH 22, Telnet 23.

- A certificate proves the server's identity and was issued by a trusted authority. This section needs internet access because it connects to a real website.

12. Apache and Nginx

Apache and Nginx are the two most common web servers. `apache2 -v` and `nginx -v` print the installed version, and `systemctl status` shows whether the service is running. These may report that the program is not installed on Kali, which is normal.

The lab's web server is Nginx running in a container. `docker logs protocol-lab-web` shows its logs, including each request you made with curl and wget.

- Checking the logs after sending a request is a good way to see both sides of a connection.

13. FTP active and passive modes

In active mode the server opens the data connection back to the client. In passive mode the client opens it to the server. Passive mode usually works better when the client is behind a firewall or NAT, because the client makes all connections.

Inside FTP, `passive` toggles the mode and `status` confirms it. `ascii` and `binary` switch the file transfer type as described above.

14. Final checklist

The checklist at the end of the README covers the main skills: finding your IP and open ports, scanning with Nmap, using HTTP, FTP and SSH, understanding mail protocols, and inspecting HTTPS certificates. Tick each item once you can do it without notes.

A good test is to explain, out loud, what the client sent and what the server replied for each protocol.

Quick reference: protocols and ports

Protocol	Standard port	Lab port	Encrypted?	Purpose
HTTP	80	8080	No	Web pages
HTTPS	443	n/a	Yes	Secure web pages
FTP	21	2121	No	File transfer
SSH	22	2222	Yes	Secure remote shell
Telnet	23	2323 (demo)	No	Remote shell (legacy)
SMTP	25	n/a	No (by default)	Sending email
POP3	110	n/a	No (by default)	Downloading email
IMAP	143	n/a	No (by default)	Synced mailboxes